

CONTENT

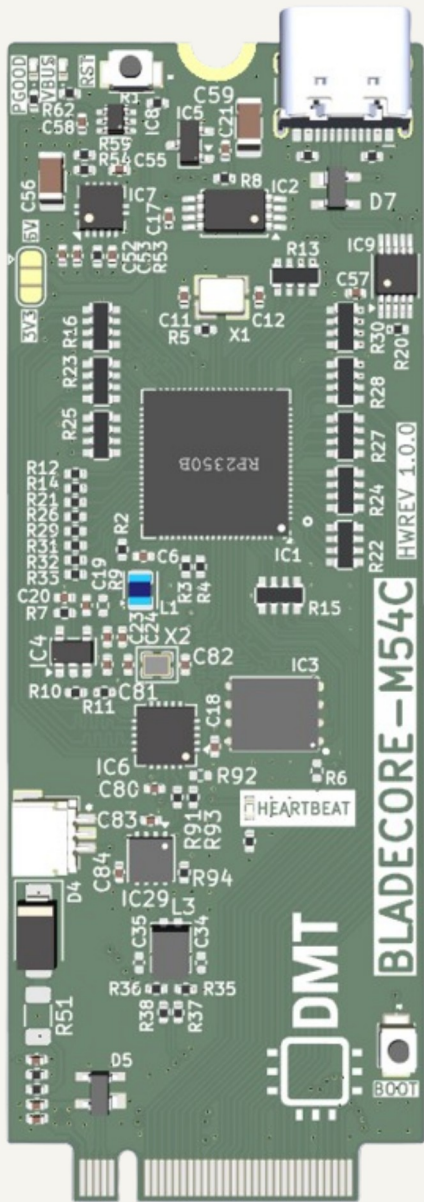
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COMMENT GUIDELINES

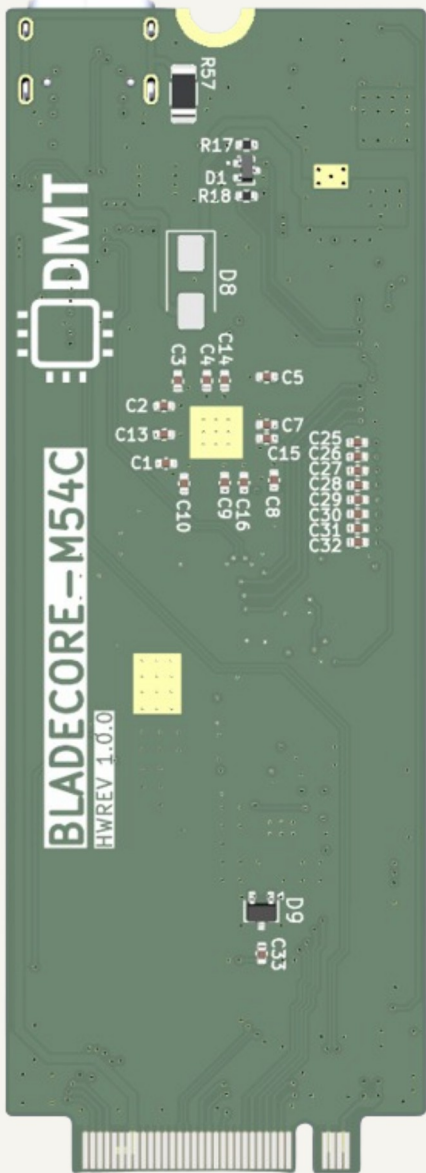
General comments are black, 50 mil size
Design notes and guidelines are blue, 50 mil size
Layout instructions are red, 50 mil size

PCB PREVIEW

TOP VIEW



BOTTOM VIEW



NOTES

Not fitted components are marked as X

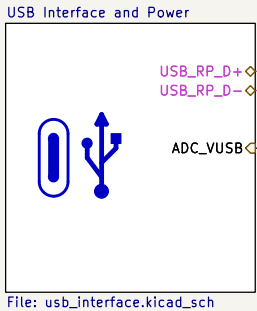
 Date: 2026-01-31 Sheet Title: Root	Board Name: BladeCore		Project Name: M54	
	File Name: BladeCore-M54C.kicad_sch		Revision: 1.0.0	Variant: C
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 1 of 9

The diagram illustrates the RP2354B USB interface circuit. The central component is the RP2354B microcontroller, which is connected to various external components and interfaces. On the left, a USB Interface block is connected to the RP2354B's ONBOARD USB and CARD EDGE USB pins. A +5V supply is connected to the USB Interface, and a +3.3V supply is connected to the RP2354B's SUPPLY pin. A USB MUX switch is connected to the ONBOARD USB and CARD EDGE USB pins. On the right, a USB Interface block is connected to the RP2354B's SUPPLY pin. A +3.3V supply is connected to the USB Interface, and a +5V supply is connected to the RP2354B's SUPPLY pin. A USB MUX switch is connected to the SUPPLY pin. The RP2354B is also connected to various other components: GPIO0-27 and GPIO37-39 to an IO block; GPIO28-31 and GPIO34-35 to an SPI block; GPIO32-33 to an I2C block; ADC AVDD to a 3.0V VREF block; GPIO47 to a GPIO36 block; GPIO46 to a GPIO40-45 block; and SWD to an SWD HEADER block. The RP2354B is also connected to a 2280 SIZE M.2 M-KEY CARD. The RP2354B is connected to a 256KB EEPROM block via I2C, a W25Q128JVPIQ 8MB FLASH block via QSPI, and a HEARTBEAT LED block via IO. The RP2354B is also connected to an MCP2515T CAN Controller block via SPI, a TCAN1044 CAN Transceiver block via CAN, and a CAN_H/CAN_L block via CAN. The RP2354B is also connected to an ADC0-ADC5 block via ADC.

Input voltage:	3.3V - 5.5 V
Absolut maximum current consumption	~200 mA
Typical current consumption	60mA
IO voltage	3.3V
Onboard LDO output capacity	1.5A
LDO Dropout	60mV

 DMT Date: 2026-01-31	Board Name: BladeCore		Project Name: M54	
	File Name: Block Diagram.kicad_sch		Revision: 1.0.0	Variant: C
Sheet Title: Block Diagram	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 2 of 9

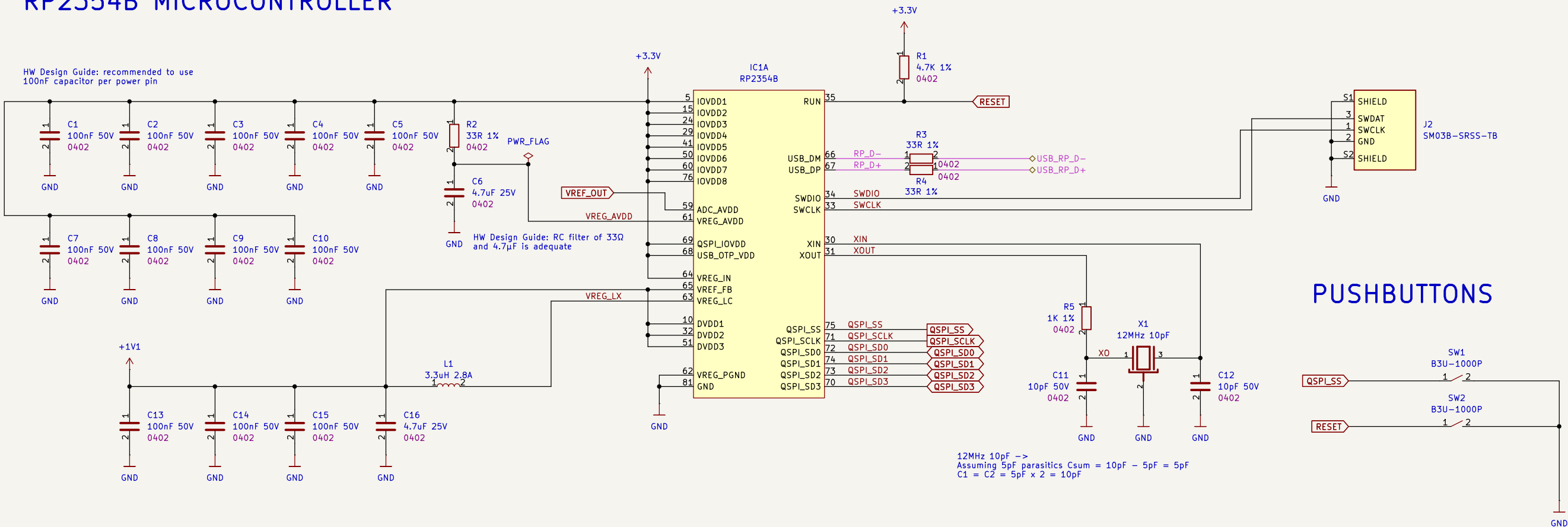
[3] Project Architecture



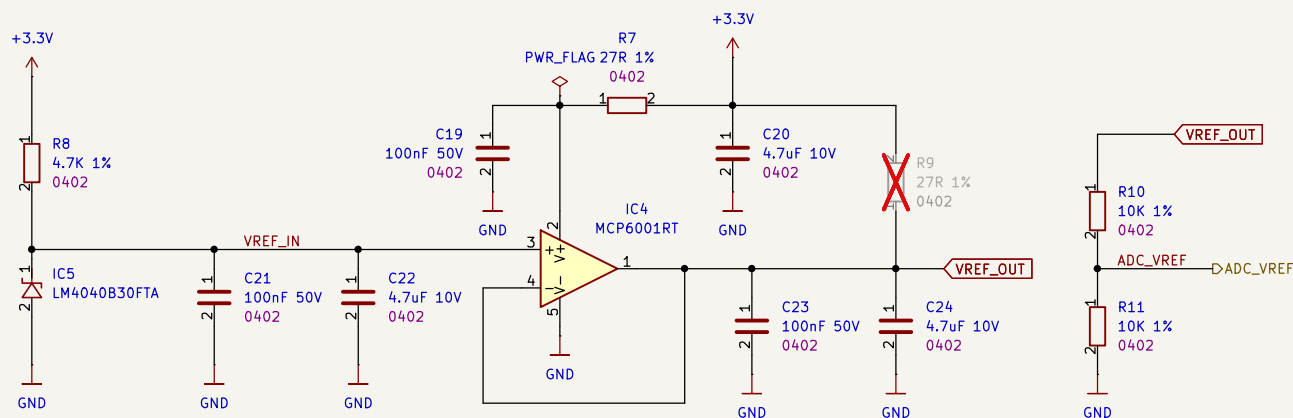
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	File Name: Project Architecture.kicad_sch		Revision: 1.0.0	Variant: C
Sheet Title: Project Architecture	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 3 of 9

[4] Microcontroller Peripherals

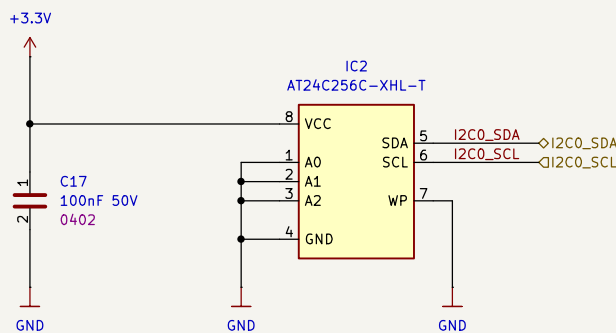
RP2354B MICROCONTROLLER



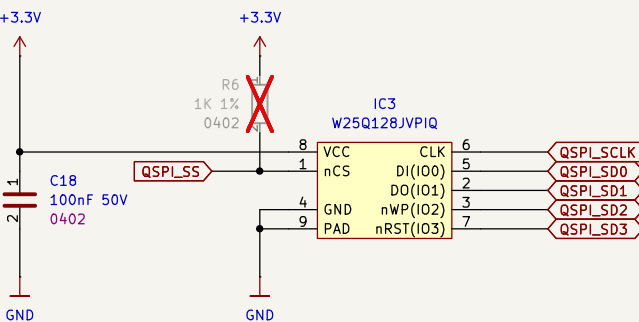
3.0V ADC VOLTAGE REFERENCE



EXTERNAL EEPROM



EXTERNAL FLASH



Date: 2026-01-31

Sheet Title:
Microcontroller Peripherals

Board Name:
BladeCore

File Name:
microcontroller_peripherals.kicad_sch

Company:
DvidMakesThings

Designer:
David Sipos
Reviewer:

Project Name:
M54

Revision:
1.0.0

Size:
A3

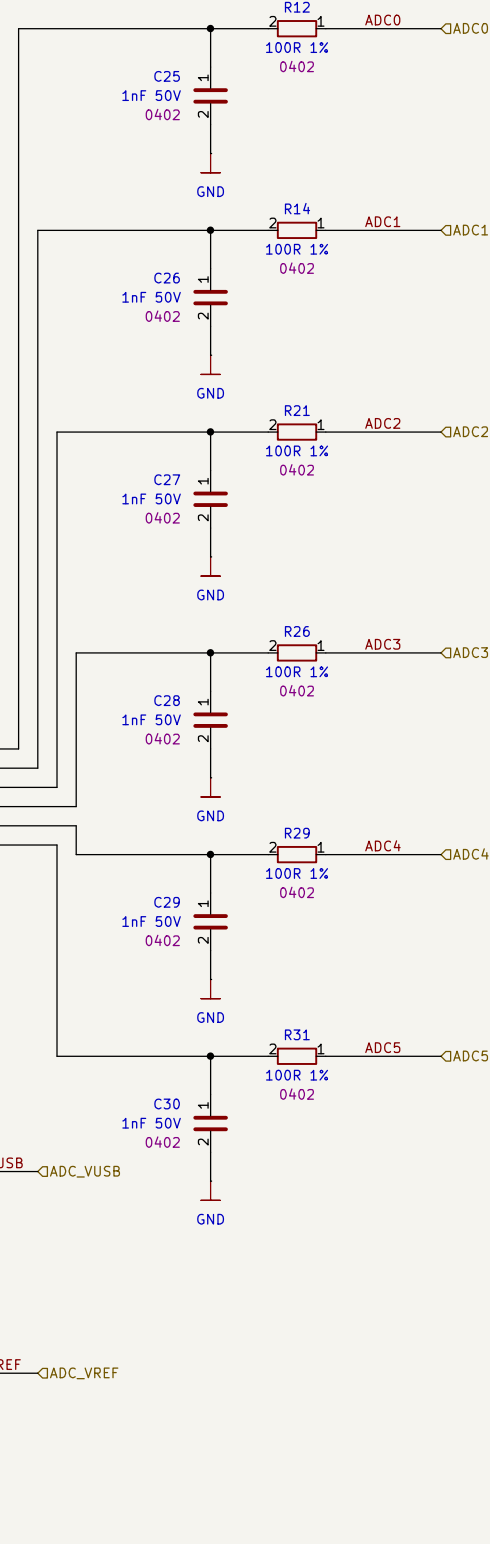
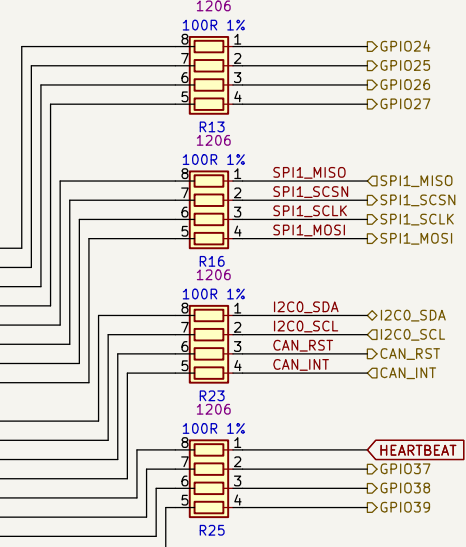
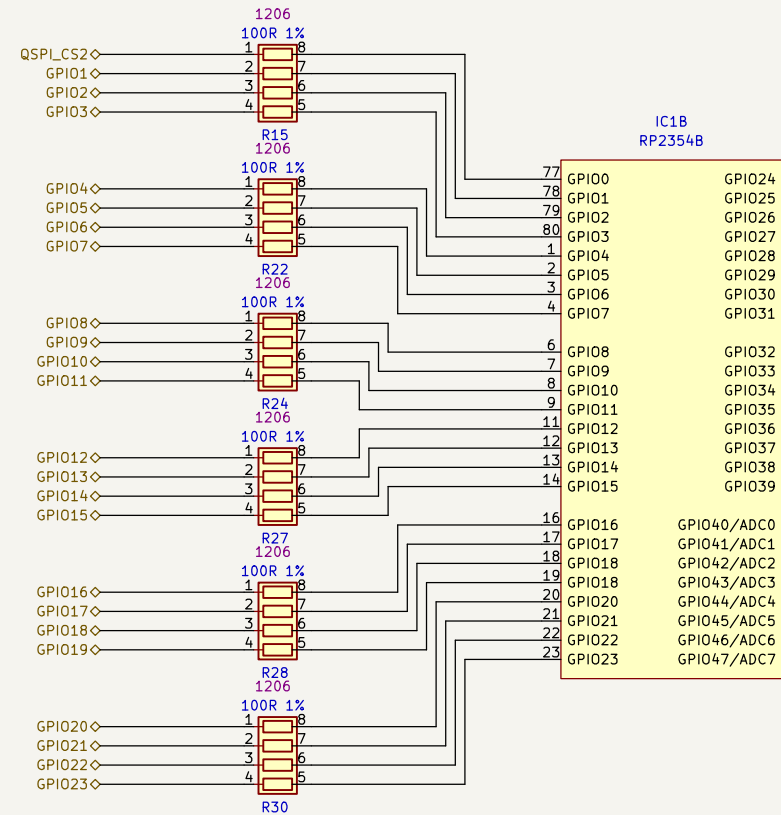
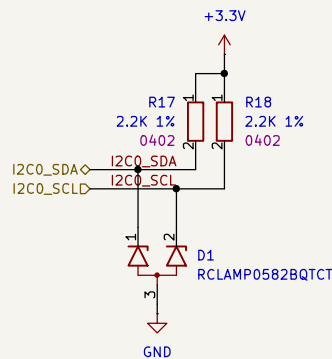
Variant:
C

Sheet:
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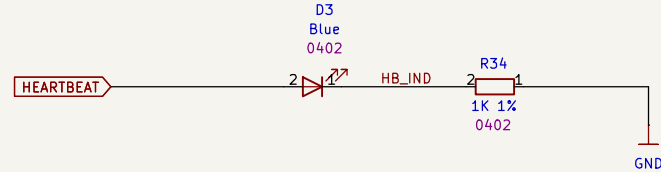
[5] Microcontroller

RP2354B GPIO ASSIGNMENT

I2C PULLUP



HEARTBEAT LED

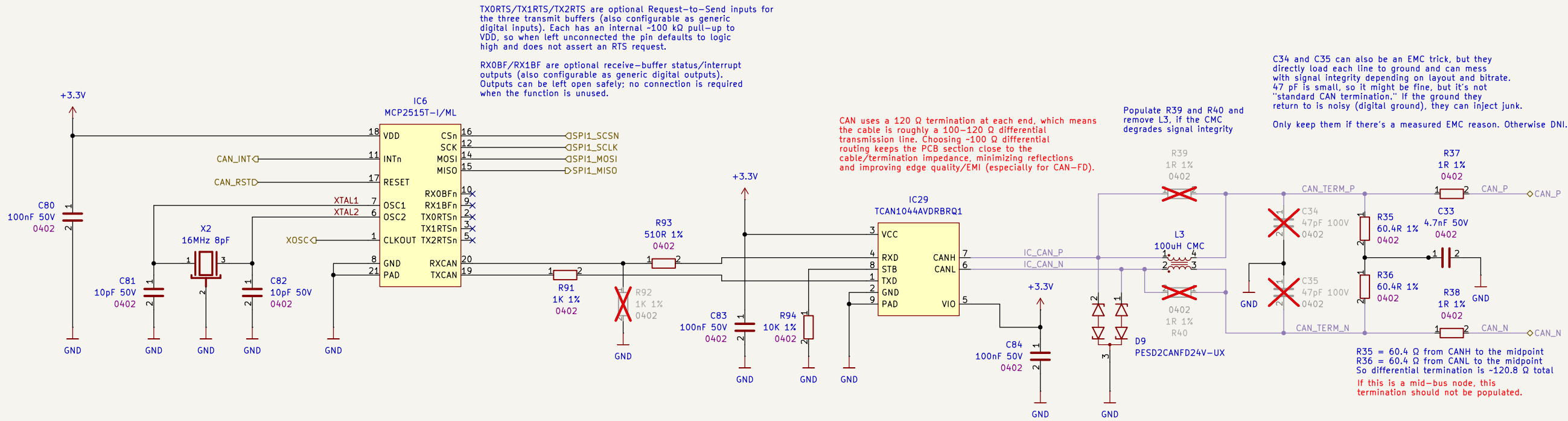


 Date: 2026-01-31 Sheet Title: Microcontroller	Board Name: BladeCore		Project Name: M54	
	File Name: microcontroller.kicad_sch		Revision: 1.0.0	Variant: C
	Company: DvidMakesThings		Designer: David Sipos Reviewer:	Size: A3
				Sheet: 5 of 9

[6] CAN Interface

MCP2515 CAN CONTROLLER

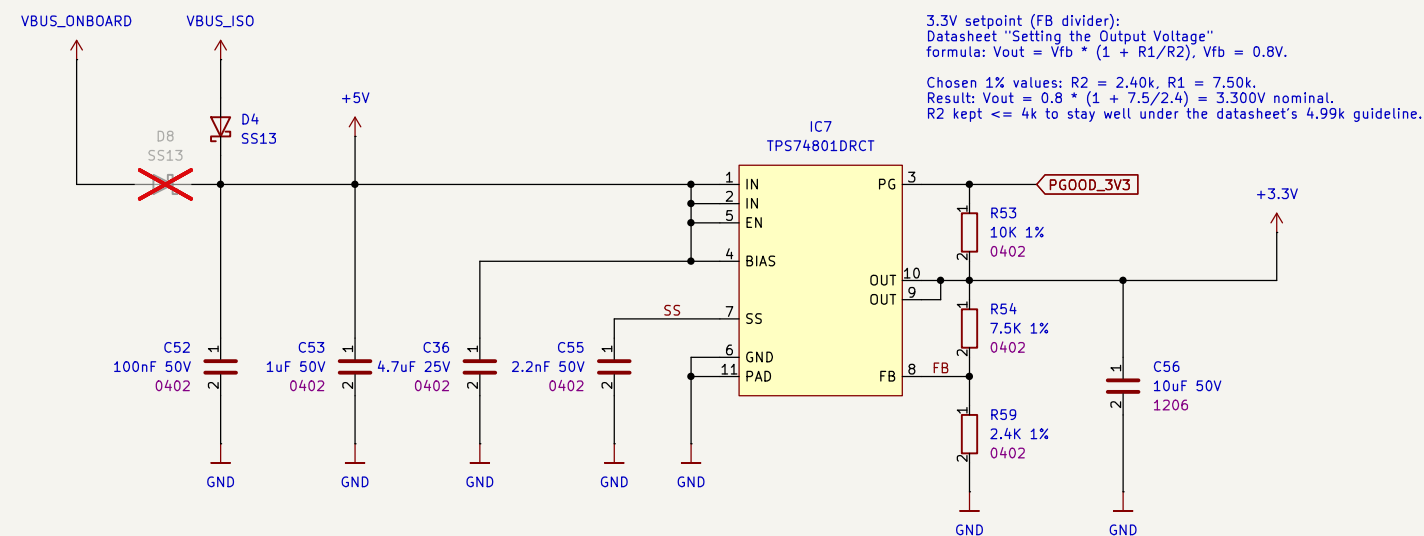
TCAN1044 CAN TRANSCIEVER



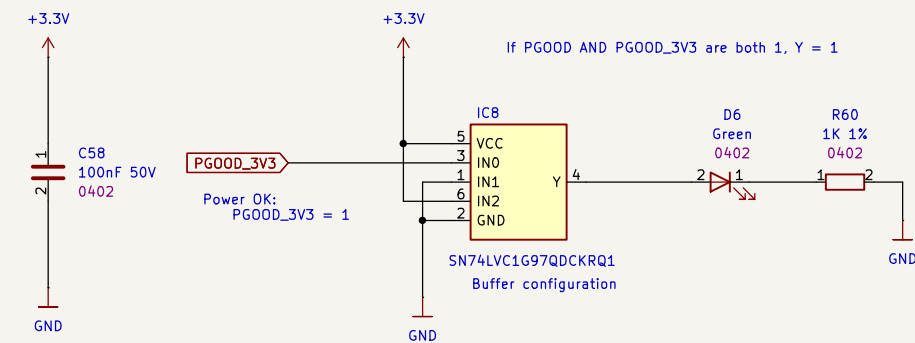
 Date: 2026-01-31 Sheet Title: CAN Interface	Board Name: BladeCore		Project Name: M54	
	File Name: can_interface.kicad_sch		Revision: 1.0.0	Variant: C
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3	Sheet: 6 of 9

[7] USB Interface and Power

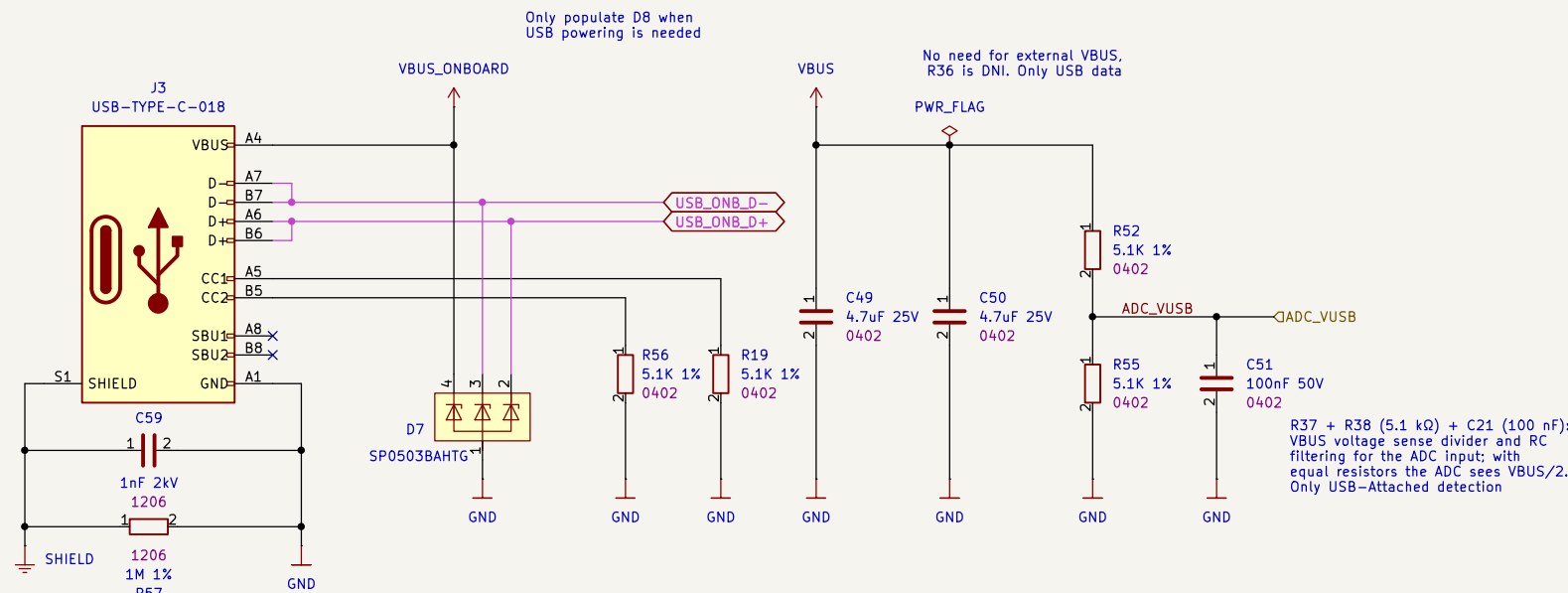
+5V to 3.3V PMIC



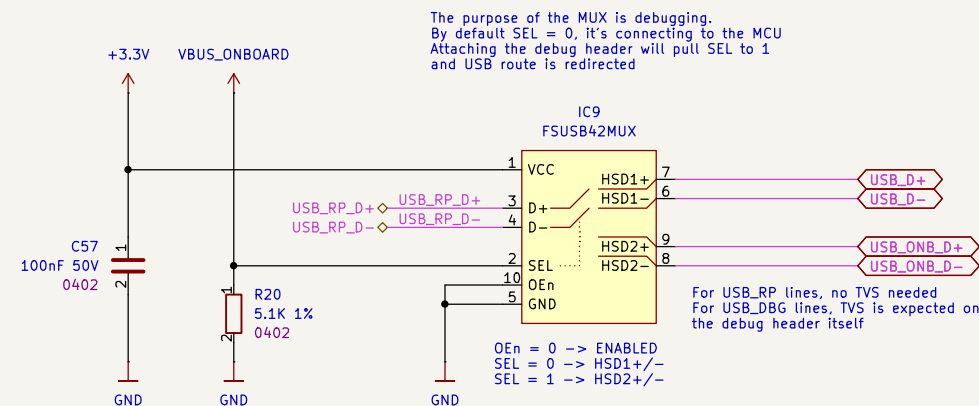
PGOOD LOGIC



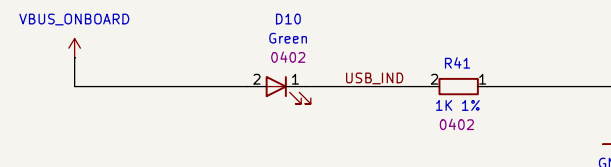
ONBOARD USB CONNECTOR



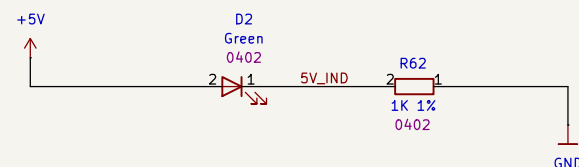
USB DATA-PATH SELECTOR



ONBOARD USB INDICATOR LED

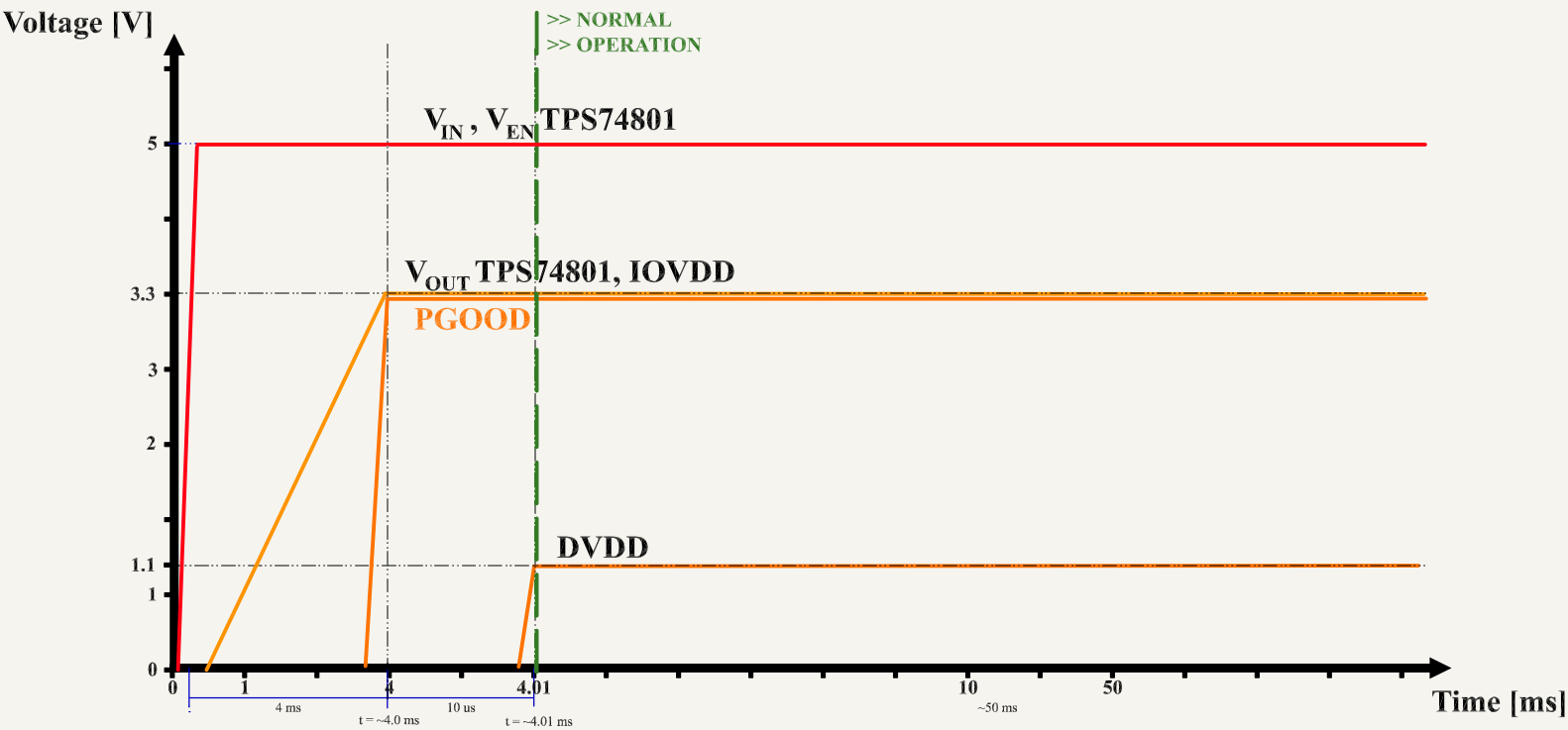


+5V INDICATOR LED



 Date: 2026-01-31 Sheet Title: USB Interface and Power	Board Name: BladeCore	Project Name: M54	
	File Name: usb_interface.kicad_sch	Revision: 1.0.0	Variant: C
	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3 Sheet: 7 of 9

[8] Power - Sequencing



#	Stage	Δt	Total elapsed
1	VIN applied (5.0V)	0	0.000 ms
2	VEN high (TPS74801 EN)	0	0.000 ms
3	VOUT = 3.3V (soft-start)	4.000 ms	4.000 ms
4	PGOOD valid	0	4.000 ms
5	RP2354B 3.3V present	0	4.000 ms
6	RP2354B 1.1V valid (POR)	0.010 ms	4.010 ms

[9] Revision History

DATE	REVISION	RESPONSIBLE	CHANGE
31.01.2026	1.0.0	DMT	INITIAL CREATION

 Date: 2026-01-31	Board Name: BladeCore		Project Name: M54	
	File Name: Revision History.kicad_sch		Revision: 1.0.0	Variant: C
	Sheet Title: Revision History	Company: DvidMakesThings	Designer: David Sipos Reviewer:	Size: A3 Sheet: 9 of 9